RA-HP 450 Axial Piston Pump

Operating and Installation Manual, Revision 3.2, March 2026
Rheintal Antriebstechnik GmbH

1. General information

The RA-HP 450 is a variable displacement axial piston pump for open circuit hydraulic systems in industrial presses, injection moulding machines and marine deck equipment. This manual applies to all RA-HP 450 variants with serial numbers starting with HP450-26. Keep this manual close to the machine for the whole service life of the pump.

1.1 Intended use

The pump is intended exclusively for pumping hydraulic fluids of the groups HLP (DIN 51524 part 2) and HEES biodegradable esters. Use with water, fuels or food liquids is not permitted. Any other use is considered improper and voids the warranty.

1.2 Safety symbols

Symbol	Meaning
DANGER	Immediate hazard which will result in death or serious injury.
WARNING	Potentially hazardous situation which could result in serious injury.
CAUTION	Situation which could result in minor injury or property damage.
NOTICE	Information to prevent damage to the pump or the installation.

2. Technical data

Parameter	Value	Unit
Nominal displacement	45	cm³/rev
Maximum displacement	50	cm³/rev
Nominal pressure	280	bar
Peak pressure (max. 1 s)	350	bar
Speed range	500 to 3000	rpm
Maximum inlet pressure (absolute)	0.8 to 30	bar
Weight (without fluid)	38	kg
Permissible fluid temperature	-20 to +80	°C
Viscosity range	10 to 400	mm²/s
Required cleanliness class	ISO 4406 20/18/15	-

The optimum operating viscosity is between 16 and 36 mm²/s. Operation below 10 mm²/s causes accelerated wear of the slipper pads and is only permitted for short periods during start-up.

2.1 Noise emission

At 1500 rpm and 250 bar the sound pressure level measured according to DIN 45635 is 72 dB(A). At 3000 rpm the level rises to 79 dB(A). Hearing protection is mandatory when working near the running pump for more than 30 minutes per day.

3. Installation

3.1 Mounting position

The pump can be installed in any position. If the shaft points upwards, the case drain line must leave the pump from the highest drain port so that the housing stays completely filled with fluid. The case drain line must end below the fluid level in the tank at all times to avoid draining the housing when the machine is stopped.

3.2 Tightening torques

Connection	Thread	Torque (Nm)
Mounting flange bolts	M12 x 1.75, strength class 10.9	110
Pressure port flange SAE 3/4"	M10	57
Suction port flange SAE 1 1/2"	M12	110
Case drain port L1 / L2	G 3/4	160
Pressure controller cover	M6	10

3.3 Commissioning

Before the first start, fill the pump housing through the upper case drain port with filtered hydraulic fluid. Start the drive at the lowest possible speed, not above 500 rpm, and run the pump for at least 2 minutes at zero load. Only then increase the pressure in steps of 50 bar up to the operating pressure. Bleed the pressure line at the highest point until fluid emerges without bubbles.

NOTICE: Never run the pump dry, not even for a few seconds. Dry running destroys the slipper pads within seconds and is not covered by the warranty.

4. Pressure controller DR

The pressure controller limits the maximum system pressure by reducing the swivel angle of the pump when the set pressure is reached. The factory setting is 250 bar. The setting range is 50 to 280 bar. One full turn of the adjustment screw changes the pressure by approximately 55 bar. Turning clockwise increases the pressure. After adjustment, secure the screw with the lock nut and tighten the nut to 10 Nm.

5. Maintenance intervals

Interval	Activity
Daily	Check fluid level, listen for unusual noise, look for external leakage.
Weekly	Check the filter clogging indicator and the fluid temperature.
Every 500 operating hours	Take an oil sample and check cleanliness class.
Every 2000 operating hours or 12 months	Replace the suction and return filters.
Every 4000 operating hours	Check the shaft seal and the mounting bolt torques.
Every 12000 operating hours	Major overhaul by RA service or an authorised partner.

The shaft seal is a wear part. A leakage of up to 3 drops per hour at the shaft seal is normal and does not require any action.

6. Troubleshooting

Symptom	Possible cause	Remedy
Pump does not deliver flow	Wrong direction of rotation	Compare the arrow on the housing with the motor rotation and swap two phases.
Loud knocking noise, foam in the tank	Air in the suction line (cavitation)	Check suction line for leaks, reduce suction height, check fluid level.
Pressure does not reach the set value	Pressure controller worn or wrongly set	Readjust controller, replace the controller cartridge if the setting drifts.
Fluid temperature above 80 °C	Cooler dirty or pressure relief valve opening permanently	Clean the cooler, check the relief valve setting (must be above the DR setting).
Leakage at the shaft, more than 3 drops per hour	Shaft seal worn or case pressure above 2 bar	Check the case drain line for restrictions, replace the shaft seal.

7. Spare parts

Part number	Description	Notes
HP450-SK-01	Seal kit complete	Includes shaft seal and all O-rings
HP450-DR-250	Pressure controller cartridge DR	Pre-set to 250 bar
HP450-SP-09	Slipper pad set (9 pieces)	Replace only as a complete set
HP450-BR-02	Bearing set	Requires press tool HP-T-11

Technical support: service@rheintal-antriebstechnik.example, +49 7621 555 0450, Monday to Friday 7:00 to 17:00 CET.